

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seven Semester of B. Tech. (CL) Examination

November - 2013

CL 401 Estimation, Costing & Financial Concepts - I

Date: 14.11.2013, Thursday

Time: 10:00 ^{c/m} p.m. To 1:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What are the different types of approximate cost estimates? Define suitability of each. 04
- (b) What is Administrative Approval? List the kind of documents that should be attached to it. 03
- (c) List the factors affecting Detail estimate of a project. 02

- Q - 2 (a) Any Contract document consists of two types of conditions of contract, General and Special. List the general conditions of contract. 05
- (b) Define: Tender, Earnest money, Security deposit, Work order and Retention money. 05

OR

- Q - 2 (a) To undertake works of project, Owner and Construction firm joined together to satisfy interest of each other through legal document is called Contract. List various contract systems. Write one in detail which is generally adopted for various construction projects. 05
- (b) Describe the particulars of Tender Notice. Who is having full authority of awarding the tender of any cost? 05

- Q - 3 (a) State rules of measurements for brick masonry work and describe the deduction criteria for Brick masonry work. 03
- (b) What degree of accuracy should be adopted for measurement of following items? 03
- i. Wood work
 - ii. Plastering work
 - iii. Reinforce concrete work
- (c) Detail specification is part of contract document. State the purpose of detail specification. 04
- (d) Write detail specification of D.P.C. work. 06

OR

- Q - 3 (a) State rules of measurements for plastering work and describe the deduction criteria for it. 03

- (b) State measurement unit of following items. 03
- Pointing
 - Lintels over opening
 - Piece work
- (c) Write general specifications of second class building. 04
- (d) Write detail specifications for cement plastering. 06

SECTION – II

- Q - 4 (a) Prepare material statement for given quantity of work. 07

Item of work	Quantity	Unit
First class brickwork in 1:6 cements mortar in superstructure in 20 cm wall.	70	Cu.m
R.C.C. work 1:2:4 excluding steel reinforcement and its bending but including centering, shuttering and binding steel. % steel of 1.2.	28	Cu.m
12 mm plastering with 1:6 cement mortar on walls	750	Sq.m

- Q - 5 (a) Workout the rate analysis of 1st class brickwork in CM (1:6) up to plinth using modular bricks – for 10 cu m. 05
- (b) Estimate quantity of R.C.C (M20) work and steel of circular column; refer section shown in Figure 1. 10

OR

- Q - 5 (a) Workout the rate analysis of 12 cm R.C.C. slab proportion (1:2:4) with 1% steel reinforcement for 10 sq m. 05
- (b) Estimate quantity of R.C.C work and steel of Slab; refer section shown in Figure 2. 10
- Q - 6 Estimate quantity of Any THREE of following for building plan shown in Figure 3. 13
- Brick work Above P.L in CM 1:6
 - Plastering work CM 1:4, 12 mm thick
 - Quantity of R.C.C work
 - Flooring work of 2.5 cm cement tiles over 7.5 lime concrete.

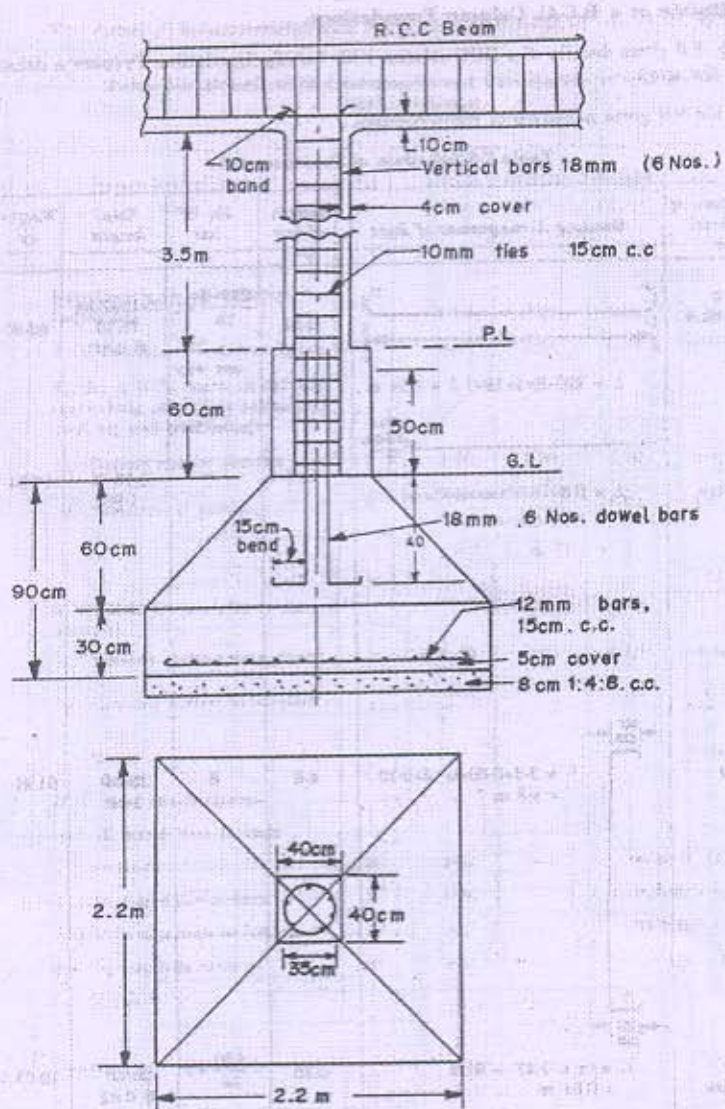
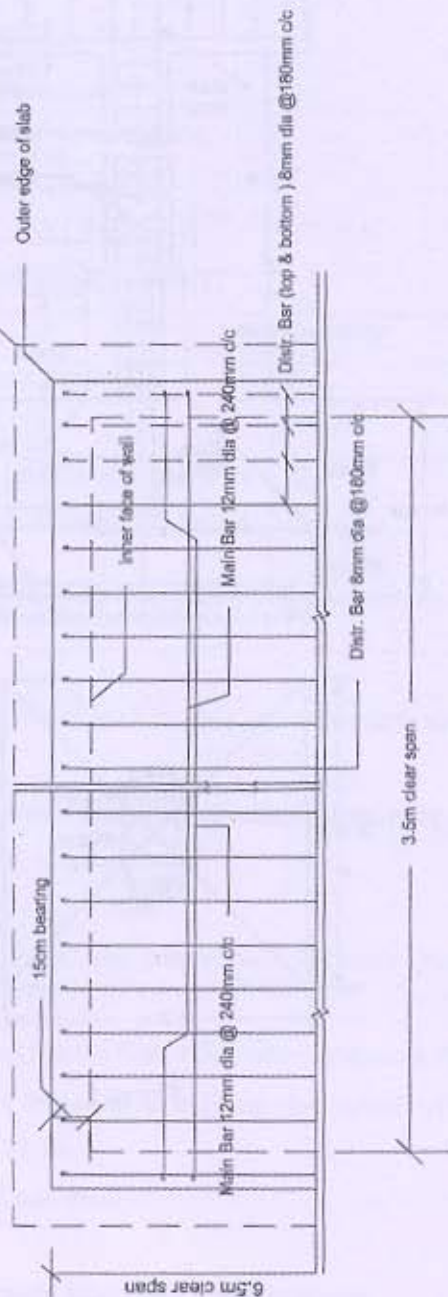


Figure 1

R.C.C ROOF SLAB



CROSS SECTION



PART PLAN SHOWING ARRANGEMENT OF REINFORCEMENT BARS

Figure 2

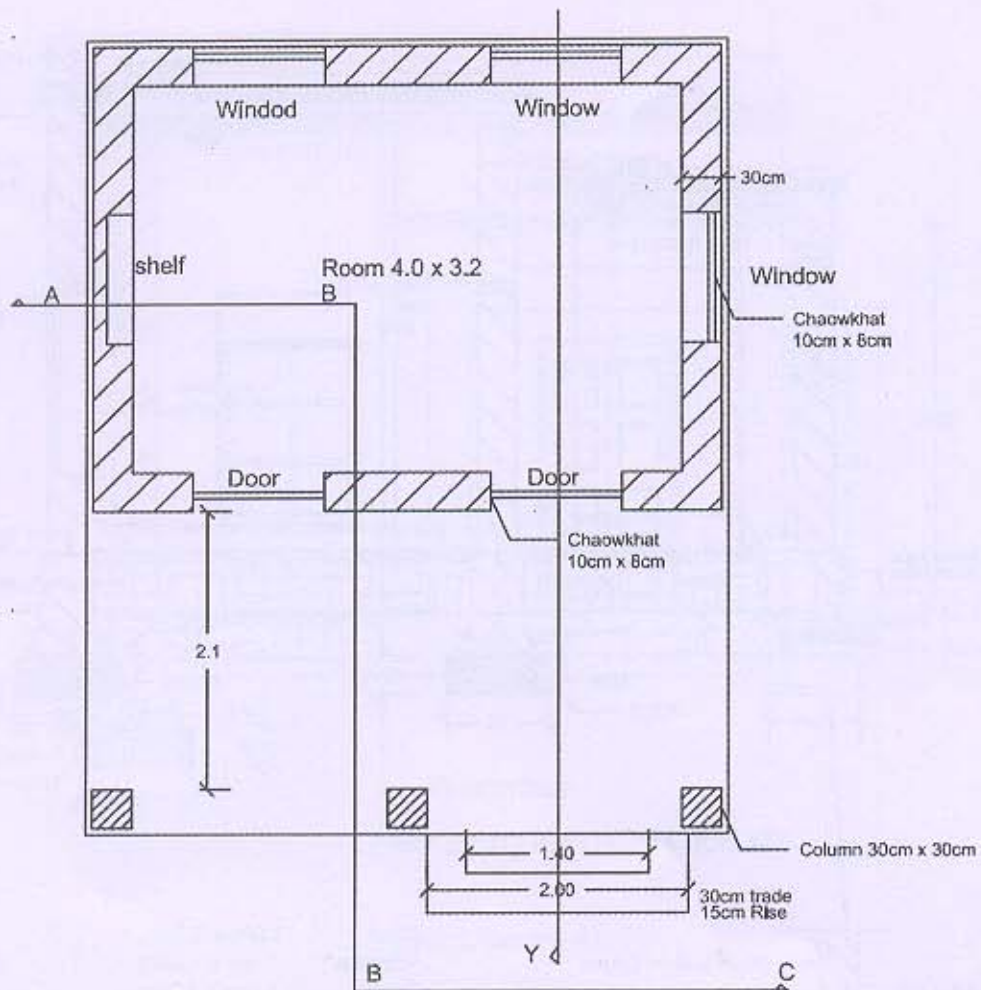
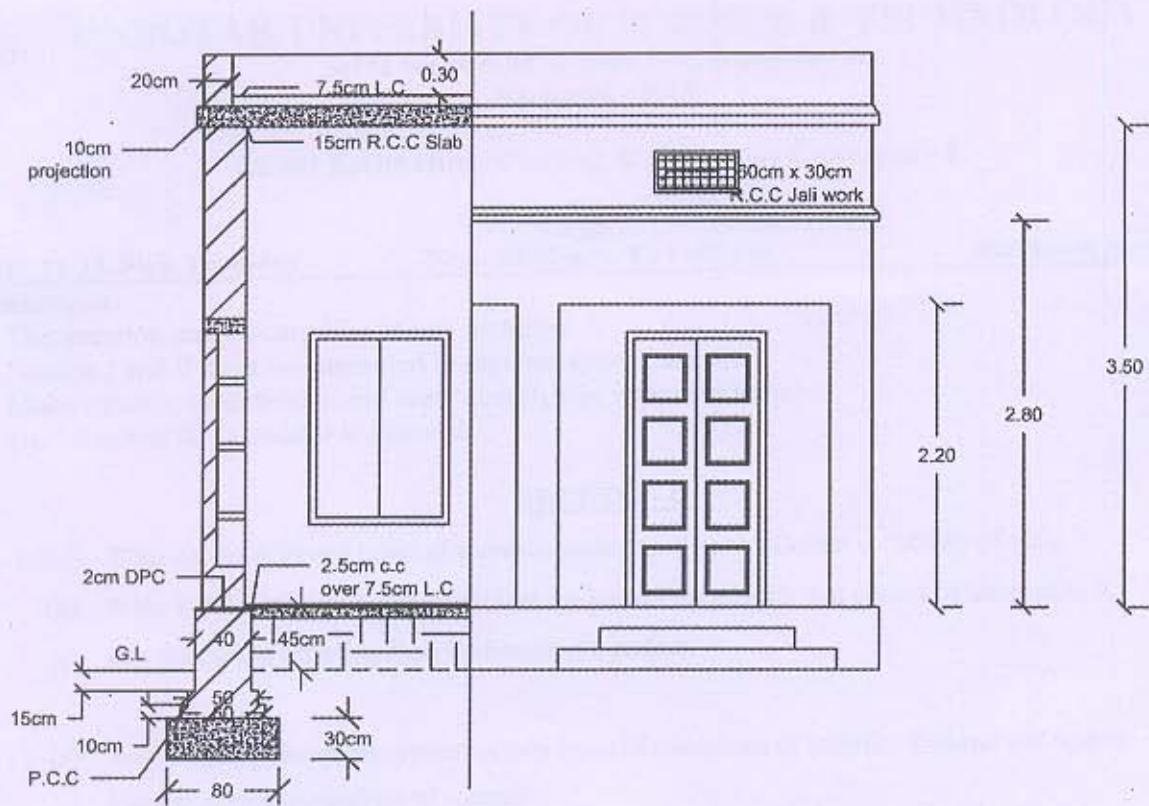


Figure 3. One room Plan



PART SECTIONAL ELEVATION ABC

Figure 5. Section ABC